CS 599 A1: Assignment 6

Due: Thursday, April 30, 2026

Total: 100 pts

Instructor: Ankush Das

- This assignment is due midnight on the above date and must be submitted electronically on Gradescope.
- You are provided a tex file, named asgn6.tex. It contains environments called solution to enter your solutions. And please include your name and BU ID in the author section (above).
- Although not recommended, you can submit handwritten answers scanned as a PDF and clearly legible.

1 Commutativity and Associativity

A binary operator $*$ is defined to be commutative if $A * B$ and $B * A$ are equivalent. In other words, $A * B \Longrightarrow B * A$ and $B * A \Longrightarrow A * B$.
A binary operator $*$ is defined to be associative if $A * (B * C)$ and $(A * B) * C$ are equivalent. In other words, $A * (B * C) \Longrightarrow (A * B) * C$ and $(A * B) * C \Longrightarrow A * (B * C)$.

Problem 1 (Mandatory; No LLMs; 10 pts) *Refer the rules in Appendix A. Which operators in ordered logic are commutative? For each operator $*$ that is commutative, provide the proofs of equivalence of $A * B$ and $B * A$, i.e., $A * B \Longrightarrow B * A$ and $B * A \Longrightarrow A * B$.*

Problem 2 (Mandatory; No LLMs; 20 pts) *Refer the rules in Appendix A. Which operators in ordered logic are associative? For each operator $*$ that is associative, provide the proofs of equivalence of $A * (B * C)$ and $(A * B) * C$, i.e., $A * (B * C) \Longrightarrow (A * B) * C$ and $(A * B) * C \Longrightarrow A * (B * C)$.*

2 Invertibility

In this problem, you need to determine which inference rules in ordered logic are invertible. Recall the definition of an invertible inference rule: if the conclusion of the rule is provable, then all the premises must be provable. In other words, given a derivation of the conclusion of a rule, you can construct a derivation of each premise of the rule.

Problem 3 (Mandatory; No LLMs; 40 pts) *Which inference rules in Appendix A are invertible? For each rule that is invertible, provide a proof of its invertibility.*

Hint: *Use the cut admissibility theorem.*

3 New Operators

Problem 4 (Mandatory; No LLMs; 10 pts) *Suppose we change the $\backslash L$ to the following:*

$$\frac{\Omega_L\ B\ \Omega_R \Longrightarrow C}{\Omega_L\ A\ (A \backslash B)\ \Omega_R \Longrightarrow C}\ \backslash L^*$$

Instead of allowing an arbitrary Ω to the left of $(A \backslash B)$, we require that the proposition on the left of $(A \backslash B)$ must be A. Based on this, answer the following questions:

1. *Does this violate identity expansion? If yes, provide a brief justification. If not, provide a derivation of identity expansion.*

2. *Does this violate cut admissibility? If yes, provide a brief justification. If not, provide a derivation of cut admissibility.*

3. *Give an example of a proposition that is derivable in ordered logic but no longer derivable if we use* $\backslash L^*$ *instead of* $\backslash L$.

Problem 5 (Mandatory; No LLMs; 10 pts) *We define a new operator* $A \diamond B$ *with the following left and right rules:*

$$\frac{\Omega_L \Longrightarrow A \qquad \Omega_R \Longrightarrow B}{\Omega_L\ \Omega\ \Omega_R \Longrightarrow A \diamond B}\ \diamond R \qquad\qquad \frac{\Omega_L\ A\ \Omega\ B\ \Omega_R \Longrightarrow C}{\Omega_L\ (A \diamond B)\ \Omega_R \Longrightarrow C}\ \diamond L$$

Does this operator satisfy identity expansion? Does it satisfy cut admissibility? In each case, provide a proof if the property holds, and a brief justification if it does not.

Problem 6 (Mandatory; No LLMs; 10 pts) *We define a new operator* $A \Diamond B \Box C$ *with the following right rule:*

$$\frac{\Omega_1\ \Omega_2 \Longrightarrow A \qquad \Omega_1 \Longrightarrow B \qquad \Omega_2 \Longrightarrow C}{\Omega_1\ \Omega_2 \Longrightarrow A \Diamond B \Box C}\ \Diamond\Box R$$

Design appropriate left rule(s) for $A \Diamond B \Box C$ *such that identity expansion and cut admissibility hold for this operator. You do not need to show the proofs of these properties, but please provide a brief justification.*

4 Optional Problems

Problem 7 (Optional; No LLMs; 20 pts) *Prove identity expansion (Theorem 1) in ordered logic.*

Problem 8 (Optional; No LLMs; 30 pts) *Provide a (partial) proof of cut admissibility (Theorem 2) in ordered logic. You only need to show all the principal and commutative cases of / operator (over).*

A Ordered Logic Inference Rules

Fuse ($\bullet$)

$$\frac{\Omega_L \Longrightarrow A \qquad \Omega_R \Longrightarrow B}{\Omega_L\ \Omega_R \Longrightarrow A \bullet B}\ \bullet R \qquad\qquad \frac{\Omega_L\ A\ B\ \Omega_R \Longrightarrow C}{\Omega_L\ (A \bullet B)\ \Omega_R \Longrightarrow C}\ \bullet L$$

Under ($\backslash$)

$$\frac{A\ \Omega \Longrightarrow B}{\Omega \Longrightarrow A \backslash B}\ \backslash R \qquad\qquad \frac{\Omega \Longrightarrow A \qquad \Omega_L\ B\ \Omega_R \Longrightarrow C}{\Omega_L\ \Omega\ (A \backslash B)\ \Omega_R \Longrightarrow C}\ \backslash L$$

Over ($/$)

$$\frac{\Omega\ A \Longrightarrow B}{\Omega \Longrightarrow B / A}\ /R \qquad\qquad \frac{\Omega \Longrightarrow A \qquad \Omega_L\ B\ \Omega_R \Longrightarrow C}{\Omega_L\ (B / A)\ \Omega\ \Omega_R \Longrightarrow C}\ /L$$

With ($\&$)

$$\frac{\Omega \Longrightarrow A \qquad \Omega \Longrightarrow B}{\Omega \Longrightarrow A \,\&\, B}\ \&R \qquad \frac{\Omega_L\ A\ \Omega_R \Longrightarrow C}{\Omega_L\ (A \,\&\, B)\ \Omega_R \Longrightarrow C}\ \&L_1 \qquad \frac{\Omega_L\ B\ \Omega_R \Longrightarrow C}{\Omega_L\ (A \,\&\, B)\ \Omega_R \Longrightarrow C}\ \&L_2$$

Plus ($\oplus$)

$$\frac{\Omega \Longrightarrow A}{\Omega \Longrightarrow A \oplus B}\ \oplus R_1 \qquad \frac{\Omega \Longrightarrow B}{\Omega \Longrightarrow A \oplus B}\ \oplus R_2 \qquad \frac{\Omega_L\ A\ \Omega_R \Longrightarrow C \qquad \Omega_L\ B\ \Omega_R \Longrightarrow C}{\Omega_L\ (A \oplus B)\ \Omega_R \Longrightarrow C}\ \oplus L$$

Unit (1)

$$\frac{}{\cdot \Longrightarrow 1}\ 1R \qquad\qquad \frac{\Omega_L\ \Omega_R \Longrightarrow C}{\Omega_L\ 1\ \Omega_R \Longrightarrow C}\ 1L$$

Identity

$$\frac{}{p \Longrightarrow p}\ \textsc{Id}$$

Theorem 1 (Identity Expansion) *For an arbitrary proposition A, $A \Longrightarrow A$ is derivable.*

Theorem 2 (Cut Admissibility) *Given derivations of $\Omega \Longrightarrow A$ and $\Omega_L\ A\ \Omega_R \Longrightarrow C$, we can construct a derivation for $\Omega_L\ \Omega\ \Omega_R \Longrightarrow C$.*